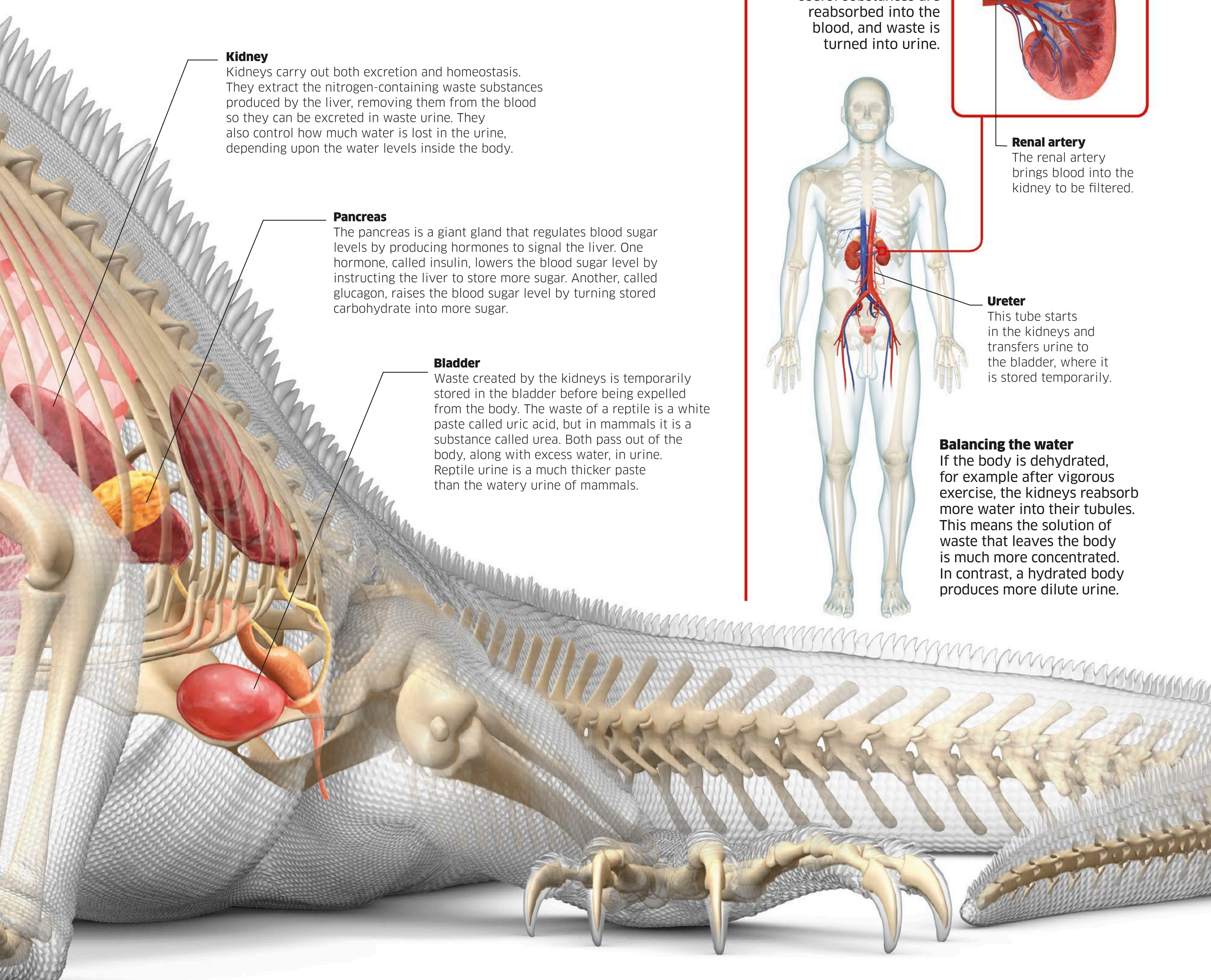
99 °F (37°C)—**the ideal temperature for enzymes** in the human body to function.

500 chemical reactions are performed **by human liver cells** in order to carry out its everyday functions.

Balancing the body

While external conditions may change from rain to shine, the internal environment of an animal's body is carefully controlled to ensure the vital processes of life can take place.

This balancing act is called homeostasis. Complex vertebrate (backboned) animals have especially good systems of homeostasis that regulate factors such as body temperature, blood sugar levels, and water levels. Alongside this, other areas of the body carry out a process called excretion to remove waste, which can be harmful if left to accumulate. This continual regulation gives the body the right set of conditions to carry out all the functions of life, such as processing food and releasing energy.



Kidney
Kidneys carry out both excretion and homeostasis. They extract the nitrogen-containing waste substances produced by the liver, removing them from the blood so they can be excreted in waste urine. They also control how much water is lost in the urine, depending upon the water levels inside the body.

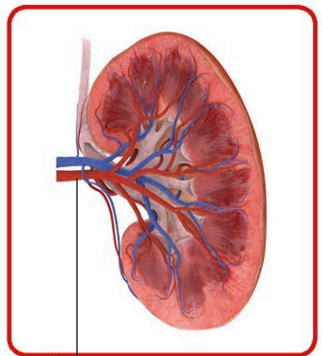
Pancreas
The pancreas is a giant gland that regulates blood sugar levels by producing hormones to signal the liver. One hormone, called insulin, lowers the blood sugar level by instructing the liver to store more sugar. Another, called glucagon, raises the blood sugar level by turning stored carbohydrate into more sugar.

Bladder
Waste created by the kidneys is temporarily stored in the bladder before being expelled from the body. The waste of a reptile is a white paste called uric acid, but in mammals it is a substance called urea. Both pass out of the body, along with excess water, in urine. Reptile urine is a much thicker paste than the watery urine of mammals.

Urinary system

Although mammal and reptile kidneys differ in their shape, both contain a complex system of blood vessels and tubes to filter the blood of waste products. These, along with the bladder, make up the urinary system. As well as removing excess water, most kidneys can also excrete in urine any unwanted salt, unlike those of the marine iguana.

Filtering the blood
Kidneys filter liquid, containing waste, directly from the blood. This liquid drains through tiny tubes, called tubules. Any useful substances are reabsorbed into the blood, and waste is turned into urine.



Renal artery
The renal artery brings blood into the kidney to be filtered.

Ureter
This tube starts in the kidneys and transfers urine to the bladder, where it is stored temporarily.

Balancing the water
If the body is dehydrated, for example after vigorous exercise, the kidneys reabsorb more water into their tubules. This means the solution of waste that leaves the body is much more concentrated. In contrast, a hydrated body produces more dilute urine.